Supplementary material

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Description. Replications on mammalian and plant allometry testing whether extrapolation failure and model-ranking inversion persist outside tokamak confinement data.

For: "Validation protocol reverses machine-learning model rankings in tokamak energy-confinement scaling" (L. Salcedo).

This document reports the two replications outside fusion referred to in the Discussion of the main text. They are supporting replications rather than evidence the main confinement result depends on: they show that the extrapolation failure measured on the ITPA database is not specific to tokamaks, and they probe what the ranking inversion needs and the extrapolation failure does not. Section and equation numbers prefixed S refer to this document; unprefixed section numbers refer to the main text.

S1 The same audit on a scaling law from another science

Every result in the main text rests on ITPA data, which leaves open whether any of it is a fact about fusion specifically. Mammalian basal metabolic rate against body mass is the same object and the older one: Kleiber [1] found in 1932 that metabolic rate scales as mass to the 3/4. Whether the true exponent is 3/4 or 2/3 has been argued for decades and we take no position; 3/4 is used here as the canonical historical reference law a fitted model has to beat. Taxonomic *order* is the unit a species belongs to, as a tokamak is the unit a discharge belongs to, and *body mass* is the axis a next instance lies beyond, as machine size is. The data is Supplement 1 of a metabolic-rate compilation deposited on Figshare (article 3549807), pinned by SHA-256 as the HDB5 files are, giving 541 species records across 11 orders whose median masses span 342x. The analysis runs through the domain-agnostic module released with this work rather than a copy of the fusion pipeline, so it is the procedure a reader would apply to their own data, and the problem has a *single predictor*: no model can win by finding a better combination of features, so the only thing separating them is functional form. Refitting the exponent freely gives 0.687 against Kleiber's 0.75, so the constraint is not free here either.

Table S1 reports the three splits and Figure S1 shows them.

The extrapolation failure reproduces. Both tree ensembles are worse than both power laws at all 8 mass cuts, and the random forest loses to Kleiber on 9 of 11 held-out orders. Two signatures reproduce with it. Tree error rises with extrapolation distance, at $\rho = +0.64$ and $+0.65$ against $+0.39$ and $+0.45$ for the power laws, which is the same sign

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Table S1: Kleiber's law under the same three splits. Scores are log-RMSE.

Model	CV, by species	leave-one-order-out	widest mass cut
Kleiber, exponent 3/4	0.396	0.440	**0.374**
power law, free exponent	0.374	0.432	0.496
hist grad. boosting	0.418	0.487	0.889
random forest	0.437	0.517	0.710

as the distance diagnostic in the main text but a much narrower contrast, since here the power laws rise with distance too. And the training-range bound binds at every cut.

The ranking reversal does not, and that bounds the claim. The trees never win the easy split either: under cross-validation by species they score 0.437 and 0.418 against the free power law's 0.374. With no inflated margin there is nothing available to invert. The HDB5 gain came from a flexible model exploiting structure across nine features; with one predictor and a relationship close to a straight line in logs, a tree has far less to exploit and does not buy the interpolation win that later reverses. So the inversion tracks whether the flexible model wins interpolation first, which no single database could have shown. Whether predictor count is itself the operative variable is a separate question this comparison does not settle.

A limitation of this replication. The 11 taxonomic orders are not statistically independent: they share phylogeny, so two orders that diverged recently are more alike than two that did not, and the held-out group is therefore not a draw from a pool of independent units. Comparative biology normally handles this with phylogenetically controlled regression [4]; this audit does not. It treats orders only as held-out groups, and models no phylogenetic covariance. What that costs is any rate quoted here: the counts below establish a direction, not a frequency with which a constraint helps.

The constraint result reproduces in direction, not in strength. Kleiber's published exponent costs +0.023 under cross-validation and wins the widest cut, 0.374 against 0.496, where the held-out order is 69x heavier than the training median, which is exactly the trade the Connor–Taylor constraints make in the main text. But it is weaker, winning at 4 of 8 cuts rather than 8 of 8, decisively at the extremes and narrowly losing in the middle. That a constraint helps out of distribution replicates; that it helps reliably does not.

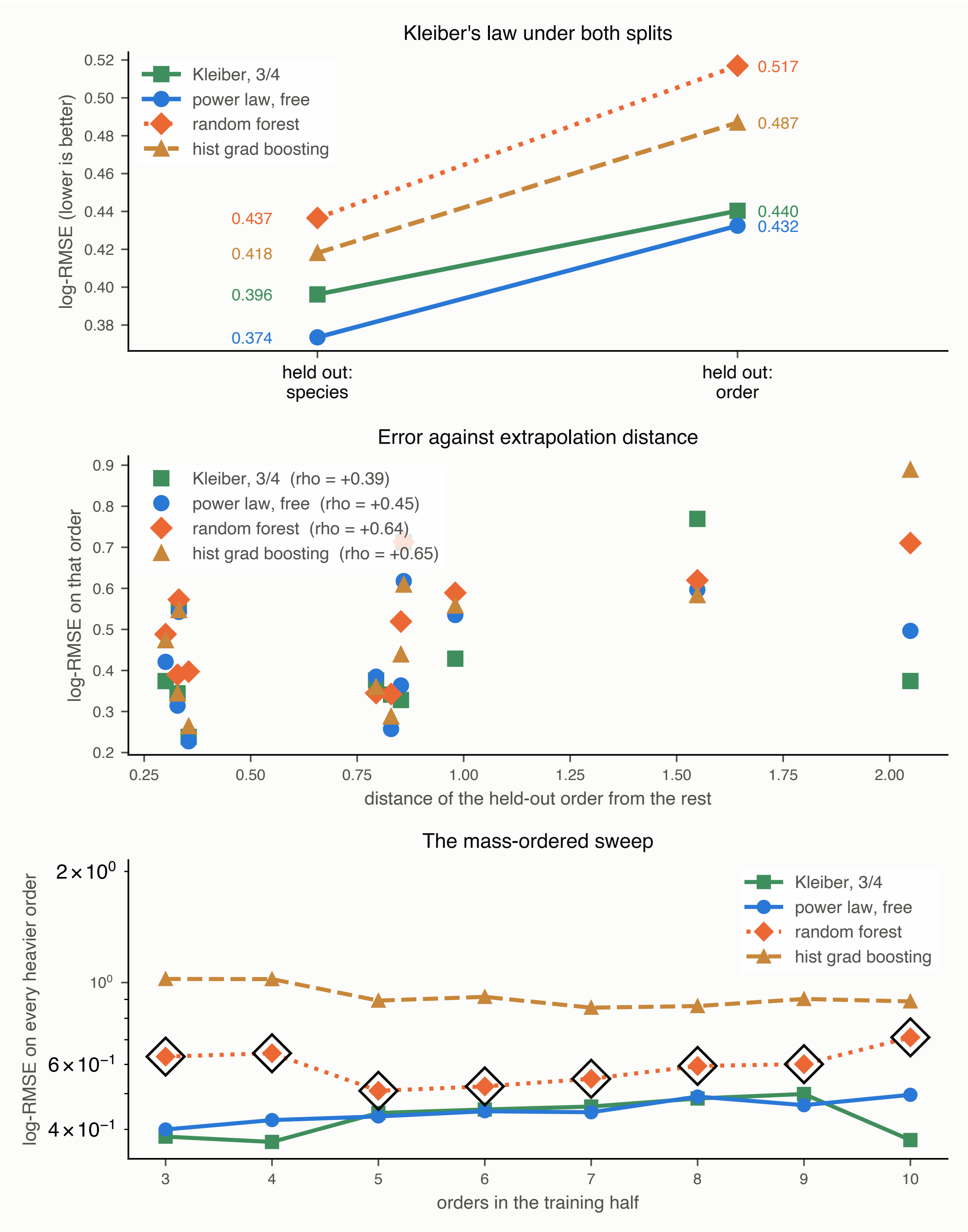

Figure S1: Kleiber's law under the same three splits. Top: no inversion, because the trees lose under both splits and so have no interpolation margin to give up. Middle: error against extrapolation distance, rising for the flexible models as it does on the ITPA H-mode database. Bottom: the mass-ordered sweep, with the circled points marking cuts where the random forest is held inside its training range.

S2 The reversal's precondition, measured directly

Sec. S1 leaves an asymmetry. The extrapolation failure reproduced in full on Kleiber's law; the ranking inversion did not, and the explanation offered there was a conjecture: with a single predictor a tree has little to exploit, never wins the interpolation split, and so has no advantage to lose. Two datasets cannot test that, because HDB5 and the mammalian compilation differ in feature count and also in science, group structure, sample size and noise.

A third dataset can probe that with a nested predictor ladder on the same rows and splits. The Biomass And Allometry Database [2] records, for individual plants, a nested set of measurements: basal diameter, height, leaf area and leaf mass. Taking the rows complete in all four gives 3599 plants across 53 species, 33 of them with the 30 individuals we require to hold a species out, spanning 36x in diameter across species medians and over seven orders of magnitude in mass. It is released CC0 and pinned by SHA-256 as the other deposits are. Species plays the part of tokamak, diameter the part of machine size, and the branching-network prediction that mass scales as diameter to the 8/3 [3] the part of IPB98(y,2): an exponent derived from theory rather than fitted, which refits here to 2.512 and costs 0.5641 against 0.5780 in sample when held.

Each rung of the ladder is a prefix of the next and all four are scored on the same rows. The rows, the splits and the model specifications are held fixed; each rung adds one named predictor to the nested feature set. Predictor count is therefore what is varied, but not the only thing that changes with it, since each added column carries its own biological information. The split structure matters and is easy to get wrong: grouped cross-validation by discharge keeps every machine on both sides, so its analogue keeps every species on both sides. Grouping the cross-validation by species instead would be a second leave-one-group-out, and against it no reversal can appear at any dimension.

Table S2 and Figure S2 separate two statements that had been travelling as one. The interpolation margin rises monotonically and crosses zero between two and three predictors. The extrapolation margin does not shrink as the trees improve; it widens, and the power law wins the held-out species at 4 of 4 rungs. The inversion therefore appears at 3 predictors and every rung above, which is exactly where the interpolation margin turns positive. What the ladder shows is that the inversion tracks the interpolation margin, not that three is a threshold: each new column adds real biological information as well as a dimension, and leaf mass in particular is closely related to the target, so the rungs differ in more than their count.

So the extrapolation failure holds in tokamaks, mammals and trees alike, at every predictor count tested here. The *inversion* is conditional, and its condition has nothing to do with the failure: it is only whether the flexible model had enough to work with to win the comparison a practitioner runs first. The inversion is therefore a symptom of the extrapolation failure rather than the failure itself. At the one- and two-predictor rungs tested here the hazard is identical, and no interpolation margin exists whose loss would signal it.

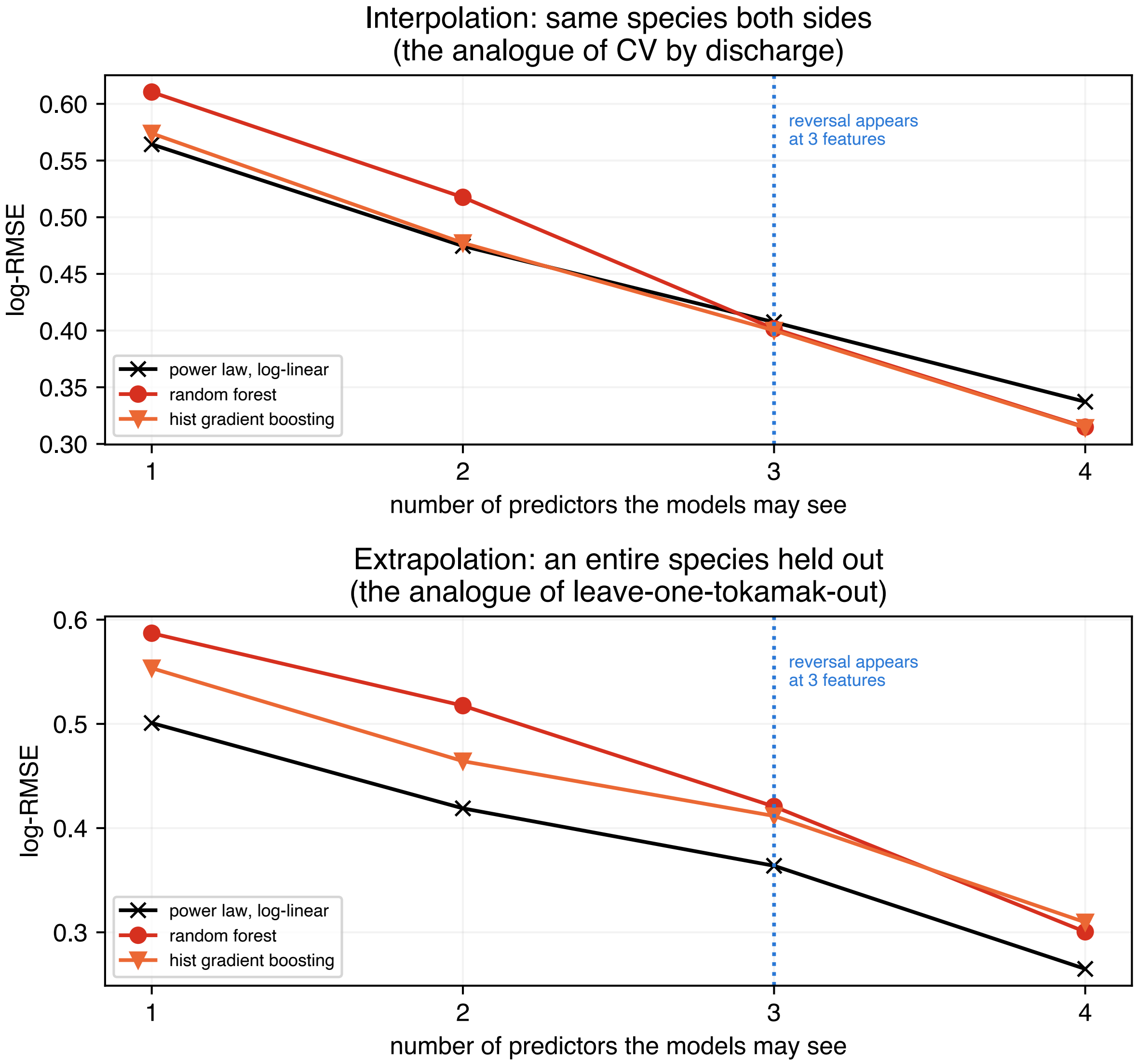

Figure S2: The reversal's precondition, on 3599 plants held fixed. Top: the interpolation split, where every species appears on both sides. Bottom: an entire species held out. The rows, splits and model specifications are held fixed; each rung adds one named predictor to the nested set. The top curves cross; the bottom ones do not.

predictors	interp. gain	held-out gain	reversal
diameter	−1.7%	−10.5%	no
+ height	−0.6%	−10.8%	no
+ leaf area	+1.8%	−13.1%	yes
+ leaf mass	+6.8%	−13.4%	yes

Table S2: The best tree ensemble's margin over the log-linear power law, on 3599 plants held fixed across the four rungs. Positive favours the trees. The interpolation column crosses zero; the extrapolation column does not, and grows slightly against them.

References

[1] M. Kleiber, "Body size and metabolism," *Hilgardia* **6**, 315 (1932). doi:10.3733/hilg.v06n11p315

[2] D.S. Falster *et al.*, "BAAD: a Biomass And Allometry Database for woody plants," *Ecology* **96**, 1445 (2015). Released CC0. doi:10.1890/14-1889.1

[3] G.B. West, J.H. Brown and B.J. Enquist, "A general model for the structure and allometry of plant vascular systems," *Nature* **400**, 664 (1999). doi:10.1038/23251

[4] J. Felsenstein, "Phylogenies and the comparative method," *Am. Nat.* **125**, 1 (1985). doi:10.1086/284325